

Aly Magassouba

Assistant Professor in Robotics and AI

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Academic Positions

2024 – present	Assistant Professor in Robotics and AI <i>University of Nottingham, School of Computer Science</i> Member of the Cyber-physical Health and Assistive Robotics Technologies (CHART) group. Research in multimodal AI and robot learning for assistive and healthcare applications. Teaching in robotics and smart sensing. Supervision of PhD students.	Nottingham, UK
2021 – 2023	Contractual Assistant Professor <i>Institut Pascal, Clermont Auvergne INP</i> Teaching in machine learning, reinforcement learning for robotics, real-time systems and virtual reality. Project management in the EU H2020 project ACROBA, with a focus on robotic perception and control. Supervision of PhD students and postdocs. PI: Prof. Youcef Mezouar.	Clermont-Ferrand, France
2017 – 2021	Research Fellow <i>National Institute of Information and Communications Technology (NICT)</i> Multimodal AI models for natural language understanding and generation for domestic service robots. PI: Prof. Komei Sugiura.	Kyoto, Japan

Education

2013 – 2016	PhD in Signal Processing and Robotics <i>Inria/IRISA, Université Rennes 1</i> Thesis: <i>Aural servo: towards an alternative approach to sound localization for robot motion control</i> . Supervisor: Dr. François Chaumette.	Rennes, France
2010 – 2012	MSc in Robotics and Intelligent Systems <i>Université Paul Sabatier</i>	Toulouse, France
2007 – 2010	BSc in Robotics and Intelligent Systems <i>Université Paul Sabatier</i>	Toulouse, France

Research

Overview	Learning to assist with AI and robots in the real world. Applications in healthcare, manufacturing and agriculture.
Communicating	Communicating with humans: bridging communication gaps with common-sense, personalisation and training support.
Perceiving	Perceiving and understanding the world: solving the ambiguities of the world with multimodal perception.
Interacting	Physically interacting with the world: interacting safely with complex objects in complex environments.

Awards and Competitions

2025	Winner, Physical Robotic Caregiving Challenge (bed-bathing track), ICRA 2025, Atlanta, USA, with the CHART RoboNotts team.
2019	Japanese Society for Artificial Intelligence Annual Conference Award.
2018	IROS RoboCup Best Paper Award, Madrid, Spain, for <i>A Multimodal Classifier Generative Adversarial Network for Carry and Place Tasks From Ambiguous Language Instructions</i> .
2018	Winner, World Robot Summit, Partner Robot Challenge (Virtual Space), Tokyo, Japan (prize of about \$100k).
2018	Japanese Society for Artificial Intelligence Award, Tokyo, Japan.
2018	Second place, RoboCup Japan Open, RoboCup@Home Simulation League, Ogaki, Japan.

Projects and Leadership

2024 – present	RoboNotts <i>University of Nottingham robotics team</i> Leading RoboNotts, the University of Nottingham's student robotics team, which competes in international robotics competitions. Winner of the Physical Robotic Caregiving Challenge (bed-bathing track) at ICRA 2025.	Nottingham, UK
ongoing	Virtual Role Simulator <i>Research project</i> A research platform for studying and training empathy in meaningful conversations, combining a large language model, an empathy-assessment framework (EPITOME), speech and 3D avatars, in workplace, education and healthcare scenarios.	
ongoing	Empathic Home <i>Research project</i> AI models of a home or building that aim to improve the well-being of its occupants.	

Teaching

2024 – present	University of Nottingham <i>School of Computer Science</i> Convenor and lecturer of <i>Introduction to Smart Things and Robotics</i> (COMP2064), second-year undergraduate module.	Nottingham, UK
2021 – 2023	Clermont Auvergne INP (SIGMA Clermont) <i>Mechanics and robotics specialisation, taught in English</i> Machine Learning; Reinforcement Learning for Robotics; Cyber-Physical Systems; Real-Time Systems; Integrated Design and Virtual Reality.	Clermont-Ferrand, France

Publications

Journal Articles

2026	Alexander Turner, Aly Magassouba , and Sobanawartiny Wijekumar. <i>Applying a Transformer-based machine-learning model to classify caregiver and infant behaviours during dyadic interactions</i> . <i>Infant Behavior and Development</i> 82 (2026), 102175. DOI: 10.1016/j.infbeh.2025.102175.	
2026	Yufeng Yang, Pei Shan Yap, Sobanawartiny Wijekumar, Aly Magassouba , and Nikhil Deshpande. <i>A Multimodal Adaptive Framework for Social Interaction with the MiRo-E Robot</i> . <i>Sensors</i> 26.4 (2026), 1209. DOI: 10.3390/s26041209.	
2024	Mélodie Daniel, Aly Magassouba , Miguel Aranda, Laurent Lequière, Juan Antonio Corrales Ramón, Roberto Iglesias Rodríguez, and Youcef Mezouar. <i>Multi Actor-Critic DDPG for Robot Action Space Decomposition: A Framework to Control Large 3D Deformation of Soft Linear Objects</i> . <i>IEEE Robotics and Automation Letters</i> 9.2 (2024), 1318–1325. DOI: 10.1109/LRA.2023.3342672.	
2021	Aly Magassouba , Komei Sugiura, and Hisashi Kawai. <i>CrossMap Transformer: A Crossmodal Masked Path Transformer Using Double Back-Translation for Vision-and-Language Navigation</i> . <i>IEEE Robotics and Automation Letters</i> 6.4 (2021), 6258–6265. DOI: 10.1109/LRA.2021.3092686.	
2021	Aly Magassouba , Komei Sugiura, Angelica Nakayama, Tsubasa Hirakawa, Takayoshi Yamashita, Hironobu Fujiyoshi, and Hisashi Kawai. <i>Predicting and attending to damaging collisions for placing everyday objects in photo-realistic simulations</i> . <i>Advanced Robotics</i> 35.12 (2021), 787–799. DOI: 10.1080/01691864.2021.1913446.	
2020	Aly Magassouba , Komei Sugiura, and Hisashi Kawai. <i>A Multimodal Target-Source Classifier With Attention Branches to Understand Ambiguous Instructions for Fetching Daily Objects</i> . <i>IEEE Robotics and Automation Letters</i> 5.2 (2020), 532–539. DOI: 10.1109/LRA.2019.2963649.	
2020	Tadashi Ogura, Aly Magassouba , Komei Sugiura, Tsubasa Hirakawa, Takayoshi Yamashita, Hironobu Fujiyoshi, and Hisashi Kawai. <i>Alleviating the Burden of Labeling: Sentence Generation by Attention Branch Encoder–Decoder Network</i> . <i>IEEE Robotics and Automation Letters</i> 5.4 (2020), 5945–5952. DOI: 10.1109/LRA.2020.3010735.	
2019	Aly Magassouba , Komei Sugiura, Anh Trinh Quoc, and Hisashi Kawai. <i>Understanding Natural Language Instructions for Fetching Daily Objects Using GAN-Based Multimodal Target–Source Classification</i> . <i>IEEE Robotics and Automation Letters</i> 4.4 (2019), 3884–3891. DOI: 10.1109/LRA.2019.2926223.	
2018	Aly Magassouba , Nancy Bertin, and François Chaumette. <i>Aural Servo: Sensor-Based Control From Robot Audition</i> . <i>IEEE Transactions on Robotics</i> 34.3 (2018), 572–585. DOI: 10.1109/TR0.2018.2805310.	
2018	Aly Magassouba , Nancy Bertin, and François Chaumette. <i>Exploiting the Distance Information of the Interaural Level Difference for Binaural Robot Motion Control</i> . <i>IEEE Robotics and Automation Letters</i> 3.3 (2018), 2048–2055. DOI: 10.1109/LRA.2018.2806560.	

- 2018 **Aly Magassouba**, Komei Sugiura, and Hisashi Kawai. *A Multimodal Classifier Generative Adversarial Network for Carry and Place Tasks From Ambiguous Language Instructions*. IEEE Robotics and Automation Letters 3.4 (2018). Presented at IROS 2018; RoboCup Best Paper Award, 3113–3120. DOI: 10.1109/LRA.2018.2849607.

Conference Papers

- 2026 Adam Biggs, Emily Burdett, **Aly Magassouba**, and Praminda Caleb-Solly. *From Terrain to Wind Noise: Contextual Factors Shaping Outdoor Robot Guide Design for Visually Impaired People*. In: Companion Proceedings of the 21st ACM/IEEE International Conference on Human-Robot Interaction (HRI). 2026, 838–843. DOI: 10.1145/3776734.3794513.
- 2026 Kohio Deflesselle, Mélodie Daniel, **Aly Magassouba**, Miguel Aranda, and Olivier Ly. *ManeuverNet: A Soft Actor-Critic Framework for Precise Maneuvering of Double-Ackermann-Steering Robots with Optimized Reward Functions*. In: IEEE International Conference on Robotics and Automation (ICRA). 2026. arXiv: 2602.14726.
- 2026 Koyo Fujii, Luis Figueredo, Praminda Caleb-Solly, Ivan Boschi, Edoardo Ida', Marco Carricato, and **Aly Magassouba**. *PhyPush: One Push is All You Need for Sensorless Physical Property Estimation with Physics-Guided Transformers*. In: IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS). 2026. arXiv: 2605.26284.
- 2026 Nishi Shishir, Aulia Nadila, **Aly Magassouba**, and Nikhil Deshpande. *Designing Intuitive Human-Drone Interfaces for Post-disaster Response*. In: Companion Proceedings of the 21st ACM/IEEE International Conference on Human-Robot Interaction (HRI). 2026, 1245–1248. DOI: 10.1145/3776734.3794606.
- 2025 Liz Dowthwaite, Karen Lancaster, Elizabeth Marsh, Emma McClaughlin, Pepita Barnard, Praminda Caleb-Solly, Harriet Cameron, Peter Craigon, **Aly Magassouba**, Frederick Moir, and Helena Webb. *An Ethical Risk Assessment of a Social Robot in the Workplace*. In: 34th IEEE International Conference on Robot and Human Interactive Communication (RO-MAN). 2025, 1859–1864. DOI: 10.1109/RO-MAN63969.2025.11217608.
- 2025 Joseph Hunt, Koyo Fujii, **Aly Magassouba**, and Praminda Caleb-Solly. *Beyond Detection – Orchestrating Human-Robot-Robot Assistance via an Internet of Robotic Things Paradigm*. In: International Conference on Social Robotics + AI (ICSR), Lecture Notes in Computer Science. 2025, 516–523. DOI: 10.1007/978-981-95-2379-5_40.
- 2020 **Aly Magassouba**, Komei Sugiura, and Hisashi Kawai. *Latent-Space Data Augmentation for Visually-Grounded Language Understanding*. In: Advances in Artificial Intelligence: Selected Papers from the Annual Conference of the Japanese Society for Artificial Intelligence (JSAI 2019), Advances in Intelligent Systems and Computing. 2020, 179–187. DOI: 10.1007/978-3-030-39878-1_17.
- 2020 **Aly Magassouba**, Komei Sugiura, and Hisashi Kawai. *Multimodal Attention Branch Network for Perspective-Free Sentence Generation*. In: Conference on Robot Learning (CoRL), PMLR 100. 2020, 76–85.
- 2016 **Aly Magassouba**, Nancy Bertin, and François Chaumette. *Audio-based robot control from interchannel level difference and absolute sound energy*. In: IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS). 2016, 1992–1999. DOI: 10.1109/IROS.2016.7759314.
- 2016 **Aly Magassouba**, Nancy Bertin, and François Chaumette. *First applications of sound-based control on a mobile robot equipped with two microphones*. In: IEEE International Conference on Robotics and Automation (ICRA). 2016, 2557–2562. DOI: 10.1109/ICRA.2016.7487411.
- 2015 **Aly Magassouba**, Nancy Bertin, and François Chaumette. *Sound-based control with two microphones*. In: IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS). 2015, 5568–5573. DOI: 10.1109/IROS.2015.7354166.
- 2011 A. Durand Petiteville, V. Cadenat, M. Courdresses, F. Delpech De Frayssinet, and **A. Magassouba**. *A comparison of several approaches to perform a vision-based long range navigation*. In: 10th International Workshop on Electronics, Control, Measurement and Signals (ECMS). 2011, 1–10.

Workshop Papers and Preprints

- 2026 **Aly Magassouba**. *Screw Attention: Rigid-Body Algebra Inside a Transformer*. 2026. arXiv: 2610.00904.
- 2026 Oussama Zaim, Mélodie Daniel, **Aly Magassouba**, Miguel Aranda, and Olivier Ly. *DRL-Based Pose Control for Double-Ackermann Robots Under Actuation Uncertainties*. In: ICRA 2026 Workshop on Uncertainty in Open-World Robotics. 2026. arXiv: 2606.00313.
- 2025 Kohio Deflesselle, Mélodie Daniel, **Aly Magassouba**, Miguel Aranda, and Olivier Ly. *Towards Safe Maneuvering of Double-Ackermann-Steering Robots with a Soft Actor-Critic Framework*. In: IROS 2025 Workshop on Safety of Intelligent and Autonomous Vehicles (SIIV-FM2L). 2025. arXiv: 2510.10332.
- 2016 **Aly Magassouba**, Nancy Bertin, and François Chaumette. *Binaural auditory interaction without HRTF for humanoid robots: A sensor-based control approach*. In: IROS 2016 Workshop on Multimodal Sensor-based Control for HRI and Soft Manipulation. 2016.

Other Publications

- 2019 **Aly Magassouba**, Komei Sugiura, and Hisashi Kawai. *A multimodal target-source classifier model for object fetching from natural language instructions*. In: Proceedings of the 33rd Annual Conference of the Japanese Society for Artificial Intelligence (JSAI). 2019, 2D3E403.

Thesis

2016

Aly Magassouba. *Aural servo: towards an alternative approach to sound localization for robot motion control.*
PhD thesis. Université Rennes 1, 2016.

Languages

French	Native.
English	Fluent.
Japanese	Intermediate.